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Do elementary flux modes combine linearly at the "atomic" level? Integrating tracer-based metabolomics data and elementary flux modes

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ABSTRACT

The elementary flux modes (EFMs) approach is an efficient computational tool to predict novel metabolic pathways. Elucidating the physiological relevance of EFMs in a particular cellular state is still an open challenge. Different methods have been presented to carry out this task. However, these methods typically use little experimental data, exploiting methodologies where an *a priori* optimization function is used to deal with the indetermination underlying metabolic networks. Available "omics" data represent an opportunity to refine current methods. In this article we discuss whether (or not) metabolomics data from isotope labeling experiments (ILEs) and EFMs can be integrated into a linear system of equations. Aside from refining current approaches to infer the physiological relevance of EFMs, this question is important for the integration of metabolomics data from ILEs into metabolic networks, which generally involve non-linear relationships. As a result of our analysis, we concluded that in general the concept of EFMs needs to be redefined at the atomic level for the modeling of ILEs. For this purpose, the concept of Elementary Carbon Modes (ECMs) is introduced.

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1. Introduction

Systems biology has recently shown that the pathway structure of cellular metabolism is much more complex than "canonical" metabolic pathways illustrated in biochemistry textbooks. The inherent plasticity of metabolic networks was described in the work of Schuster et al. (2000), based on the concept of elementary flux modes (EFMs). In essence, the EFMs approach enables us to decompose the whole metabolic network into minimal modes of behavior, which naturally links them to the traditional concept of metabolic pathways. The application of the EFMs approach has turned out to be effective in elucidating a number of relevant questions related to cellular metabolism: analysis of metabolic capabilities in different organisms (Schuster et al., 2002); metabolic network reconstruction (Liao et al., 1996); functional genomics (Dandekar et al., 1999); prediction of novel metabolic pathways (Schuster et al., 1999); analysis of maximum molar yield of biotransformations (Carlson et al., 2002); analysis of robustness and flexibility of the metabolic network (Stelling et al., 2002). For this reason EFMs analysis has become a key area of research in the post-genomic era.

EFM analysis faces now two challenges. Firstly, novel methods are needed to deal with the combinatorial explosion of EFMs in large metabolic networks. Until now, the full set of EFMs can only be computed in networks of moderate size (few hundreds of compounds and reactions). Active research is currently being accomplished in two directions here: (i) improvement of previous algorithms (Terzer and Stelling, 2008), and (ii) selection of a relevant set of EFMs via optimization, *e.g.* (de Figueiredo et al., 2009). In addition, systematic methods to analyze the vast amount of information from EFMs are also being investigated (Pérès et al., 2011). The second challenge refers to elucidating how physiologically significant are EFMs in different situations. This is important to establish novel pathways to target in a wide range of biomedical and biotechnological applications (Hellerstein, 2008).

The physiological significance of an EFM is characterized by its underlying steady-state flux, which we referred to here as *EFM activity*. Other names have been used in the literature, *e.g.* elemen-

Abbreviations: EFMs, elementary flux modes; ILEs, isotope labeling experiments; ECMs, elementary carbon modes; AMMs, atom mapping matrices; IDV, isotopomer distribution vector; IMMs, isotopomer mapping matrices; PPC, phosphoenolpyruvate carboxylase reaction; pep, phosphoenolpyruvate; oaa, oxoloacetate.

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Nomenclature

C	total number of metabolites
R	total number of reactions
K	total number of EFMs
J_k	activity of EFM k ($k = 1,\ldots, K$)
X_c	concentration of compound c ($c = 1, \ldots, C$)
S	stoichiometric matrix of dimensions $C \times R$
S_{cr}	stoichiometric coefficient of compound c ($c = 1, \ldots, C$) in reaction r ($r = 1, \ldots, R$)
v_r	flux of reaction r ($r = 1, \ldots, R$)
E	EFM matrix of dimensions $R \times K$
e_k	column k of matrix E ($k = 1, \ldots, K$) of dimensions Rx1
e_{rk}	relative flux of the reaction r ($r = 1, \ldots, R$) in EFM k ($k = 1, \ldots, K$)
a_c	number of carbons of metabolite c ($c = 1, \ldots, C$)
b_c	number of isotopomers of metabolite c ($c = 1, \ldots, C$)
$\mathrm{AMM}^{r\alpha\beta}$	atom mapping matrix (AMM) of reactant α and product β in reaction r of dimensions $a_\beta \times a_\alpha$
$\mathrm{IMM}^{r\alpha\beta}$	isotopomer mapping matrix (IMM) of reactant α and product β in reaction r of dimensions $b_\beta \times b_\alpha$
I^c	isotopomer distribution vector (IDV) of metabolite c ($c = 1, \ldots, C$) of dimensions $b_c \times 1$
I'^c	non-normalized IDV of metabolite c ($c = 1, \ldots, C$) of dimensions $b_c \times 1$
n_r	number of reactants in reaction r ($r = 1, \ldots, R$)
N^r	reactants vector of reaction r of dimensions $n_r \times 1$
N^r_γ	metabolite index as defined in S of reactant γ ($\gamma = 1, \ldots, n_r$,) in reaction r ($r = 1, \ldots, R$)
P_k	set of metabolites produced in at least one reaction in EFM k ($k = 1, \ldots, K$)
$I'(t+1)^\beta_k$	updated non-normalized IDV of metabolite β in iteration t of EFM k ($k = 1, \ldots, K$)
$I^{\beta,\mathrm{exp}}$	experimentally measured IDV of metabolite β
v_β	experimentally measured exchange flux of metabolite β
u^β_k	number of molecules of β produced by EFM k when $J_k = 1$

tary mode flux (Poolman et al., 2004), flux regulation coefficient (Nookaew et al., 2007), etc. To determine EFM activity, current methods generally build upon the previous knowledge of (part of) the steady-state reaction fluxes (Schwartz and Kanehisa, 2005; Wiback et al., 2003; Wlaschin et al., 2006; Zhao and Kurata, 2009). The set of experimental and computational tools to determine reaction fluxes are typically named "fluxomics" (Cascante and Marin, 2008). We here distinguish between exchange and intracellular fluxes. Some exchange fluxes can be easily (experimentally) measured, *e.g.* uptake rate of substrate (for example, glucose) and secretion rate of some metabolites (lactate). In contrast, intracellular fluxes are more difficult to determine and a more detailed consideration is required.

Intracellular fluxes can be computed by solving a system of linear equations with measured exchange fluxes as input data (Stephanopoulos et al., 1998). However, this problem is generally underdetermined. In order to reduce such under-determination, the use of metabolomics data from isotope labeling experiments (ILEs), is emerging as a very powerful strategy (Matsuoka and Shimizu, 2010; Sauer, 2006; Shimizu, 2002; Tang et al., 2009; Wiechert, 2001). Tracer-based metabolomics data from ILEs provide experimental labeling patterns of certain key metabolites when the in vitro culture is fed with substrates labeled with certain isotopic tracer, *e.g.* ^{13}C. Labeling distribution in such key metabolites further constrains reaction fluxes. The integration of ILEs into metabolic networks is therefore an important task in systems biology (Cascante and Marin, 2008).

In this article we discuss whether metabolomics data from isotope labeling experiments and the EFM activity can be set up as a system of linear equations. Up to our knowledge, no previous work in the literature has discussed whether (or not) EFMs and ILEs can fit into a linear framework. The objective is two-fold: (i) to refine current approaches in order to compute EFM activities by the direct use of tracer-based metabolomics data and (ii) to integrate metabolomics data from ILEs into metabolic networks in a linear fashion using the concept of elementary flux modes. The use of linear equations here is important, since it can provide better theoretical insights for ILEs, and it reduces the computational expense with respect to current methods, which use non-linear equations (Antoniewicz et al., 2007; Sauer et al., 1999; Schmidt et al., 1997; Wiechert et al., 1999; Zupke and Stephanopoulos, 1994).

2. Methods

2.1. Theory of EFMs

Schuster et al. (2000) defined an elementary flux mode (EFM) as a minimal set of enzymes that can operate at steady state with all irreversible reactions used in the appropriate direction. To illustrate this concept, consider a metabolic network that involves C compounds and R reactions. For each reaction r ($r = 1, \ldots, R$), a flux variable, v_r, is defined. The set of reactions can be divided into reversible and irreversible reactions. We here split reversible reactions into two irreversible steps, therefore:

$$v_r \geq 0, \quad r = 1, \ldots, R \tag{1}$$

The set of metabolites is partitioned into two subsets: external (Ext) and internal (I) metabolites. The steady-state assumption is typically applied to internal metabolites, therefore

$$\frac{dX_c}{dt} = \sum_{r=1}^{R} S_{cr} v_r = 0, \quad c \in I \tag{2}$$

We denote X_c the concentration of a given compound c, ($c = 1, \ldots, C$). In addition, S_{cr} represents the stoichiometric coefficient associated with compound c ($c = 1, \ldots, C$) in reaction r ($r = 1, \ldots, R$). As usual in the literature (Schilling et al., 2000; Schuster et al., 2000) substrates and products have negative and positive stoichiometric coefficients, respectively. External metabolites are typically sources/sinks, *e.g.* glucose, or certain metabolites whose modeling is not fully captured in the metabolic network under study, *e.g.* ATP.

Aside from Eqs. (1) to (2), EFMs are required to satisfy a simplicity condition. This condition was termed the non-decomposability condition (Schuster et al., 2000), which establishes that no subset of reactions of a given EFM can operate in steady state. Therefore, Eqs. (1) and (2) plus the non-decomposability condition define an EFM.

To illustrate the concept of EFMs, consider the example metabolic network in Fig. 1a, which comprises 9 reactions and 6 metabolites. This toy metabolic network represents a metabolic system involving glycolysis and a pentose phosphate pathway. The resulting set of EFMs is shown in Fig. 1b.

Note that the non-decomposability condition implies that an EFM defines a subset of reactions whose flux has one degree of freedom. This means that in EFMs reaction fluxes are proportional with respect to a scalar quantity. We denote as e_{rk} the relative flux of reaction r ($r = 1, \ldots, R$) in EFM k ($k = 1, \ldots, K$), K being the number of EFMs in a given network. We denote as E the $R \times K$ matrix, whose columns are the relative fluxes of the K EFMs (e_k, $k = 1, \ldots, K$). The E matrix in our example metabolic network is shown in Fig. 1c.

2.2. EFM Isotopomer Distribution

Isotope labeling experiments are based on the introduction of substrates labeled with isotopic tracers at specific locations. The inter-conversion of substrates via (intracellular) enzyme-catalyzed reactions presents asymmetries at the atomic level (Stephanopoulos et al., 1998). These asymmetries are found in the steady state isotopic enrichment of metabolites, whose experimental measurement allows us to infer intracellular fluxes. The isotopic enrichment of metabolites is represented with the concept of isotopomers, *i.e.* isomers with isotopic atoms. These isotopomers use different isotopic tracers. We will focus on carbon tracers, which are the most extended in the analysis of metabolic networks.

In this section we describe how to determine the isotopomer distribution of the active metabolites in an EFM, which, by some abuse of terminology, is here referred to as EFM isotopomer distribution. We show that, for certain

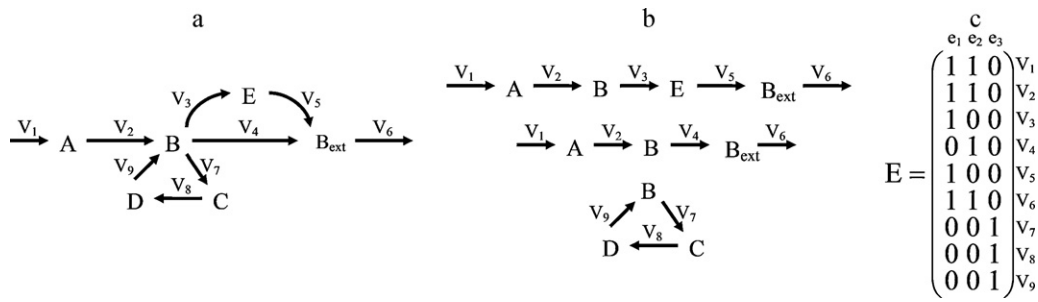

Fig. 1. (a) Example metabolic network; (b) EFMs; (c) *E* matrix.

conditions, EFM isotopomer distribution is a structural property and it can be easily calculated by slightly modifying the method presented in Schmidt et al. (1997).

As noted above, an EFM e_k characterizes a minimal set of enzyme-catalyzed reactions that can operate at steady-state, with $e_{rk} > 0$ if the reaction r is active in the EFM, $e_{rk} = 0$ otherwise ($r = 1, \ldots, R$). Each active reaction in the EFM inter-converts a number of metabolites. An active (internal) metabolite in an EFM participates in at least one active reaction in such EFM. To be able to follow the transfer of carbon flux from reactants to products in each reaction step, Atom Mapping Matrices (AMMs) were introduced by Zupke and Stephanopoulos (1994). To illustrate this concept, Fig. 2a(i) shows the phosphoenolpyruvate carboxylase reaction (PPC), which catalyzes the addition of CO_2 to phosphoenolpyruvate (pep) to form a 4-carbon metabolite oxoloacetate (oaa). An atom mapping matrix, $\mathrm{AMM}^{r\alpha\beta}$, is defined for each pair of reactant–product (α–β) in reaction r ($S_{\alpha r} < 0, S_{\beta r} > 0$), namely $\mathrm{AMM}^{r\alpha\beta}_{ij} = 1$ if the j-th carbon atom of reactant α ($j = 1, \ldots, a_\alpha$) goes to the i-th carbon atom of product β ($i = 1, \ldots, a_\beta$) in reaction r, and 0 otherwise, where a_α and a_β are the number of atoms in the reactant and product, respectively. Fig. 2a(ii) shows the AMM for CO_2 and oaa and for pep and oaa in the PPC reaction, respectively.

As noted above, isotopomers have the same molecular configuration, differing in the carbon isotopes. Therefore, the number of isotopomers for a given metabolite c is $b_c = 2^{a_c}$, where a_c is the number of atoms of compound c ($c = 1, \ldots, C$), assuming we have only two different kinds of isotopes, generally ^{12}C and ^{13}C. The work of Schmidt et al. (1997), introduced the isotopomer distribution vector (IDV), I^c, $c = 1, \ldots, C$, which defines the proportion of each isotopomer in the pool of metabolite c and satisfies that $\sum_{j=1}^{b_c} I^c_j = 1$. Fig. 2b shows an example IDV for the substrates in PPC reaction as well as the corresponding isotopomers using a dyadic notation (*i.e.* 000 is the unlabeled isotopomer, 001 an isotopomer labeled in its 1st atom, etc.).

In the work of Schmidt et al. (1997), the concept of isotopomer mapping matrices (IMMs) was also introduced. IMMs are 0/1 matrices that specify if isotopomer i of product β can be formed with isotopomer j of reactant α in reaction r ($\mathrm{IMM}^{r\alpha\beta}_{ij}$), $S_{\alpha r} < 0, S_{\beta r} > 0, i = 1, \ldots, b_\beta$ and $j = 1, \ldots, b_\alpha$. As for AMMs, we have an IMM for each pair of substrate-product in reaction r. Fig. 2c shows the IMM for pep-oaa and CO_2-oaa. As presented in Schmidt et al. (1997), IMMs are determined *in silico* starting from the knowledge of AMMs, which can be obtained from different databases (Suthers et al., 2007).

The knowledge of IMM matrices for a reaction r, in conjunction with IDVs for reactants, allows one to calculate the IDV for the products of such reaction r. Eq. (3) introduces a general formula to calculate the non-normalized IDV for product β in reaction r, I'^{β}, which defines the proportion of each isotopomer of product β without being normalized.

$$I'^{\beta} = \mathrm{IMM}^{rN^i_1\beta} I^{N^r_1} \otimes \mathrm{IMM}^{rN^i_2\beta} I^{N^r_2} \otimes (\ldots) \otimes \mathrm{IMM}^{rN^i_{n_r}\beta} I^{N^r_{n_r}} \quad \text{such that} \quad S_{\beta r} > 0 \tag{3}$$

Note that N^r represents the reactants of reaction r ($r = 1, \ldots, R$), namely N^r_γ provides the metabolite index of reactant γ ($\gamma = 1, \ldots, n_r$), as considered in the stoichiometric matrix. We denote n_r the number of reactants of reaction r. The symbol $\otimes$ represents the element-wise matrix multiplication.

With Eq. (4), IDV for product β, I^β, is obtained by normalizing I'^{β}. Fig. 2d shows how to calculate the IDV for oaa in reaction PPC, given the IDVs and IMMs previously defined in Fig. 2b and c, respectively.

$$I^{\beta} = \frac{I'^{\beta}}{\sum_{j=1}^{b_\beta} I'^{\beta}_j} \tag{4}$$

In Eq. (3) we considered only one reaction producing metabolite β. In general, we may have more than one reaction producing a metabolite β in an EFM. This fact is taken into account in Eq. (5).

$$I'^{\beta}_k = \sum_{r=1, e_{rk}>0, S_{\beta r}<0}^{R} e_{rk} [\mathrm{IMM}^{rN^r_1\beta} I^{N^r_1} \otimes \mathrm{IMM}^{rN^r_2\beta} I^{N^r_2} \otimes (\ldots) \otimes \mathrm{IMM}^{rN^r_{n_r}\beta} I^{N^r_{n_r}}], \quad \beta \in P_k \tag{5}$$

We denote as P_k the set of metabolites which participates in at least one reaction as a product in EFM k. e_{rk} ($r = 1, \ldots, R$) represents the relative flux associated with reaction r in an EFM. In essence, Eq. (5) weighs the pool of isotopomers of product β generated for each reaction r producing β ($e_r > 0$ and $S_{\beta r} > 0, r = 1, \ldots, R$) in proportion to their relative fluxes e_{rk}. As in Eq. (3), the resulting I'^{β}_k needs to be normalized using Eq. (4).

Note here that in the approach presented in Schmidt et al. (1997), and other later works, *e.g.* Wiechert et al. (1999), the term e_{rk} is substituted by the actual reaction fluxes, v_r, which are unknown and whose values need to be computed from isotopomer data of certain metabolites via stochastic optimization. Instead, for the determination of EFMs isotopomer distribution, we used e_{rk} which is fixed from EFM analysis. This is technically feasible due to the fact that EFMs form solutions (sub-networks) with one degree of freedom. This clearly simplifies the difficulty in solving Eq. (5), which does not involve any stochastic component.

Eq. (5) defines a set of non-linear equations which describe the flow of isotopomers of each metabolite in an EFM. These equations enable to determine how IDVs are redistributed in steady-state when a stationary flow of substrate entering the EFM, *e.g.* glucose, has been labeled with ^{13}C. This set of non-linear equations can be solved using a number of appropriate non-linear techniques. Here, for convenience, we have chosen to use an iterative (fixed-point) technique, which transforms Eq. (5) to the scheme shown in Eq. (6).

$$I'(t+1)^{\beta}_k = \sum_{r=1, e_{rk}>0, S_{\beta r}<0}^{R} e_{rk} [\mathrm{IMM}^{rN^r_1\beta} I(t)^{N^r_1} \otimes (\ldots) \otimes \mathrm{IMM}^{rN^r_{n_r}\beta} I(t)^{N^r_{n_r}}], \quad \beta \in P_k \tag{6}$$

Here $I'(t+1)^{\beta}_k$ represents the new value of the non-normalized IDV vector in iteration $t+1$ in EFM k. To achieve convergence, we need a sufficient number of iterations, which depends on the structure and consequently on the complexity of EFMs, *e.g.* internal cycles.

We assumed that the labeled influx of substrate is at steady-state and, therefore, its IDV is fixed to I^{in}, as observed in Eq. (7). Note that $I(t)^{in}$ indicates the value of I^{in} at the t-th iteration t.

$$I(0)^{in} = I(1)^{in} = \cdots = I(t)^{in} = I^{in} \tag{7}$$

To solve Eq. (6), which is subject to the boundary condition in Eq. (7), we assumed that all intermediate metabolites in the EFM are initially unlabeled, as shown in Eq. (8). These initial conditions are logical from biochemical point of view.

$$I(0)^c = [1, 0, 0, 0, \ldots, 0, 0, 0]^T, \quad \forall c \in D \tag{8}$$

Considering these initial and boundary conditions, Eq. (6) can be easily solved by a fixed point iterative method. A detailed example can be found in the Supplementary Material. Furthermore, for more complex problems and larger networks improved efficient algorithms for accelerated steady-state convergence (Shrof and Keller, 1993) and even for efficient optimization (Luna-Ortiz and Theodoropoulos, 2005) based on fixed-point iterations can be used.

Finally, the solving procedure for Eq. (6) subject to the boundary and initial conditions in Eqs. (7) and (8), respectively, is completely deterministic. Furthermore, our computational experience with thousands of EFMs under different initial conditions has showed that Eq. (6) is convergent for this problem. Therefore its solution for given initial conditions is unique, which shows, as mentioned previously, that the EFM isotopomer distribution is a structural property. This was expected, since, as noted above, EFMs form solutions with one degree of freedom at the flux level. Therefore, given an initial labeling input condition, the EFM isotopomer distribution is fixed and an invariant structural property of metabolic networks.

2.3. A Linear Model Linking EFMs and ILEs?

It is well-known that any possible steady-state flux distribution can be reconstructed as a linear combination of EFMs with non-negative coefficients, as reflected

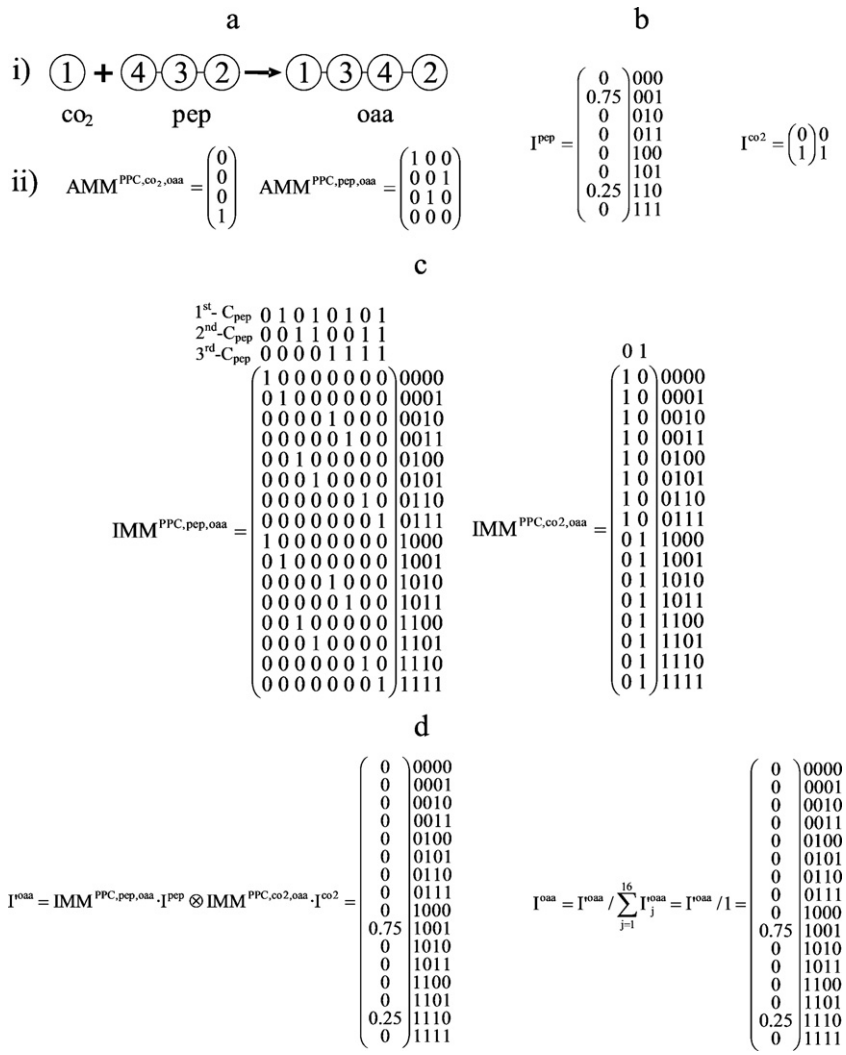

Fig. 2. (a) AMMs for phoshphoenolpyruvate carboxylase (PPC); (b) IDVs for substrates in PPC; (c) IMMs for PPC; (d) determination of IDV for products in PPC.

in Eq. (9). We denote J_k the contribution of EFM k to the steady-state flux distribution, which is referred to as the activity of EFM k.

$$v_r = \sum_{k=1}^{K} J_k e_{rk}, \quad J_k \geq 0, \quad r = 1, \ldots, R \tag{9}$$

As we showed in the previous sub-section, for the given initial and boundary conditions defined above, each EFM k defines a unique isotopomer distribution for each of its active metabolites (β), denoted I_k^{β}. Based on this, the question heading this section can be reformulated as follows: can we reconstruct the measured isotopomer distribution of a given metabolite ($I^{\beta,\exp}$) as a linear combination of the isotopomer distribution of this metabolite emerging from each EFM (I_k^{β})?

Such a linear combination is reflected in Eq. (10). J_k is precisely the activity of EFM k, as in Eq. (9), $v_{\mathrm{exc}(\beta)}$ is the exchange flux of metabolite β, which can be experimentally measured, and u_k^{β} represents the number of molecules β produced by EFM k per unit of EFM activity k, *i.e.* when $J_k = 1$. Note here that in Eq. (10) it was assumed that we have labeling data of exchange metabolites. This equation can be easily reformulated for internal metabolites.

$$v_{\mathrm{exc}}(\beta) I^{\beta,exp} = \sum_{k=1}^{K} u_k^{\beta} J_k I_k^{\beta}, \quad J_k \geq 0, \quad \forall k \in K \tag{10}$$

Eq. (10) links metabolomics data from isotope labeling experiments ($I^{\beta,\exp}$) to the EFM activity (J_k), as we posed in the introduction section. Note its analogy with Eq. (9), which reconstruct reaction fluxes with a linear combination of EFMs. Now, we need to analyze if Eq. (10) holds in general. The validity of Eq. (10) also answers the question posed in the title of this article as to whether (or not) EFMs combine linearly at the atomic level. This analysis is accomplished in Section 3.

3. Results

The question as to EFMs combine linearly at the atomic level, *i.e.* the validity of Eq. (10), is answered by means of an example. Let us consider the example metabolic network shown in Fig. 1a. Fig. 3a shows the network in Fig. 1a with some additional assumptions. Firstly, it is assumed that all metabolites in the network have three carbons. As to the input labeling conditions, it can be observed that the first carbon of metabolite A is labeled, therefore its IDV is: $I^A = [0\ 1\ 0\ 0\ 0\ 0\ 0]^T$ (see above the definition of IDV). In addition, for the sake of simplicity, we just introduced an asymmetry at the carbon level in reactions (3) and (8). In reaction (3), the first and second carbon in metabolite B are swapped when converting to metabolite E, while in reaction (8) the same occurs between first and third carbon of metabolite C when converting to metabolite D. In the rest of reactions, the carbon backbone is not affected.

With the information given above, we can apply the systematic method presented in the previous section to calculate the isotopomer distribution of active metabolites in each of the EFMs emerging from network in Fig. 3a. As introduced above, we have

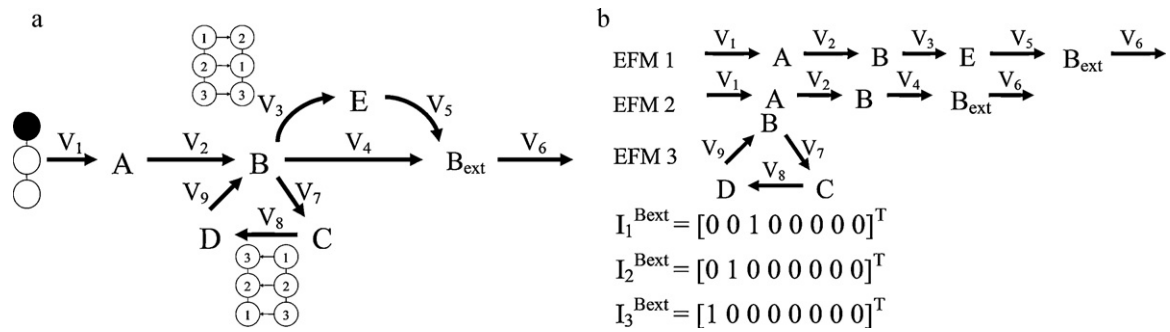

Fig. 3. (a) Atomic features of metabolic network in Fig. 1a; (b) isotopomer distribution of B_{ext} in the EFMs emerging from the network Fig. 1a with metabolite *A* having the first carbon fully labeled.

3 EFMs, which are shown again in Fig. 3b. Fig. 3b also includes the isotopomer distribution of B_{ext} for each of three EFMs (I_k^{β}, $k = 1, 2, 3$). We can observe that in the first EFM all labeled carbon is in the second position of B_{ext}, while in the second EFM is in the first position of B_{ext}. The third EFM is cyclic and, therefore, it does not involve an input of metabolite *A*. This implies that, when operating alone, no metabolite is labeled assuming Eq. (8).

To discuss whether EFMs combine linearly at the atomic level, we focused on B_{ext}. Eq. (10) was then applied to B_{ext}, as observed in Eq. (11).

$$\begin{aligned} I^{B_{ext},\exp} &= \frac{u_1^{B_{ext}} J_1 I_1^{B_{ext}} + u_2^{B_{ext}} J_2 I_2^{B_{ext}} + u_3^{B_{ext}} J_3 I_3^{B_{ext}}}{v_{exc(\beta)}} \\ &= \frac{u_1^{B_{ext}} J_1 I_1^{B_{ext}} + u_2^{B_{ext}} J_2 I_2^{B_{ext}}}{v_{exc(\beta)}} \end{aligned} \tag{11}$$

A linear combination of EFMs at the carbon level hypothesizes that the pool of isotopomers of B_{ext} is not dependent on the cyclic EFM, since $u_3^{B_{ext}} = 0$, as reflected in Eq. (11). To validate (or reject) this hypothesis, let us consider the two possible scenarios described in Table 1. In the first scenario, we assumed that the cyclic EFM supports no flux, *i.e.* $J_3 = 0$, while J_1 and J_2 were active. Then, for these fluxes, we calculate the actual labeling in B_{ext} using the method presented in Schmidt et al. (1997) (see Table 1). When the actual labeling of B_{ext} was introduced in Eq. (11) the same solution for the EFM activity was obtained. Therefore, in this case, EFMs combine linearly at the carbon level.

In the second scenario, we gave a positive flux to all EFMs, including the cyclic EFM, *i.e.* $J_3 = 1$. Using the same procedure, we calculated the actual labeling of B_{ext} and we checked whether (or not) our linear approach was able to reproduce the same fluxes. In this case, our method not only failed to describe the fluxes, but an incompatibility was found. Indeed, our linear approach cannot predict the labeling of the third carbon of B_{ext} (see actual labeling of B_{ext} in Table 1). This is due to the cyclic EFM interacts with other EFMs via metabolite *B*. These interactions cannot be captured by our linear model. Let us consider in more depth these interactions between EFMs.

We will set our focus on metabolites shared by several EFMs. In our example, metabolite *B* is used in the three EFMs. For the sake of simplicity, let us consider that the activity of the first EFM is zero, *i.e.* $J_1 = 0$, while the other EFMs are different than zero, $J_2 \neq 0$, $J_3 \neq 0$. In our linear model posed in Eq. (11) it is assumed that the isotopomers pool of metabolite *B* in the second EFM does not interact with the isotopomer pool of *B* formed in the cyclic EFM. In the analogy shown in Fig. 4a, this implies that EFMs behave as independent flux "pipes". This assumption is not true, as shown in Fig. 4b, which represents the actual behavior of the network and how the isotopomers pools of metabolite *B* of both EFMs are merged. Indeed, this merging is not represented by Eq. (11), since no EFM accounts for the exchange flux of isotopomers of metabolite B_{ext} having cycled in EFM_3 an odd number of times, which actually labels the third carbon of B_{ext}. In summary, the interaction of EFMs generates new paths at the carbon level not considered in any EFM by itself. These paths can only be modeled by introducing non-linear relationships between the activities of EFMs. For instance, Eq. (12) shows the non-linear relationship between the EFM activity and the labeling of third carbon of B_{ext} when $J_1 = 0$.

$$\text{Enrichment of third carbon of } B_{ext} = \frac{J_3}{J_2 + 2J_3} \tag{12}$$

This behavior is indeed quite common in real metabolic networks. See Supplementary Material for an example network based on a simplified model of pentose phosphate cycle and glycolysis.

3.1. Elementary Carbon Modes (ECMs)

As noted above, the interaction of EFMs generates novel paths at the atomic level not considered in the EFMs *per se*, which generally leads to inconsistencies in the proposed linear model. To be able to capture all possible atomic movements in a metabolic network, the concept of EFMs needs to be adapted at the carbon level.

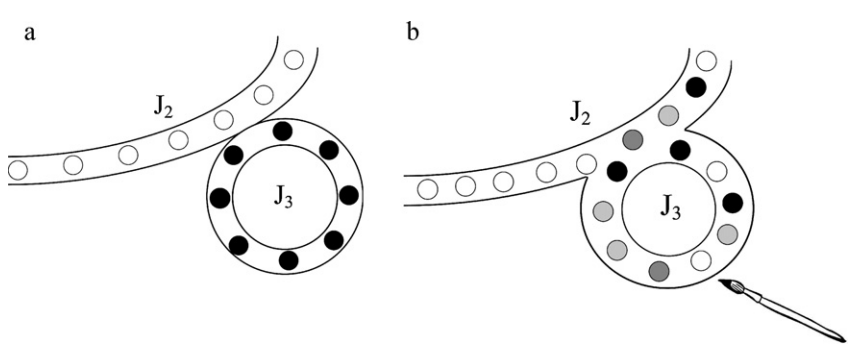

Fig. 4. Analogy between EFMs and pipelines assuming (a) a linear model; (b) a non-linear model.

Table 1
Role of the cyclic EFM in the labeling of B_{ext}.

	$J_3 = 0$	$J_3 \neq 0$
Actual fluxes	$J_1 = 1; J_2 = 2; J_3 = 0$	$J_1 = 1; J_2 = 2; J_3 = 1$
Actual labeling of B_{ext}	$I^{B_{ext},exp} = [0, 2/3, 1/3, 0, 0, 0, 0, 0]^T$	$I^{B_{ext},exp} = [0, 8/15, 4/15, 0, 1/5, 0, 0, 0]^T$
Eq. (11)	$\begin{cases} \frac{2}{3} = \frac{J_1 \cdot 0 + J_2 \cdot 1}{3} \\ \frac{1}{3} = \frac{J_1 \cdot 1 + J_2 \cdot 0}{3} \end{cases}$	$\begin{cases} \frac{8}{15} = \frac{J_1 \cdot 0 + J_2 \cdot 1}{3} \\ \frac{4}{15} = \frac{J_1 \cdot 1 + J_2 \cdot 0}{3} \\ \frac{1}{5} = \frac{J_1 \cdot 0 + J_2 \cdot 0}{3} \end{cases}$

Fig. 5a shows a similar metabolic network as that presented in Fig. 1a. Fig. 5b shows this network at the carbon level. We assumed two-carbon molecules for the sake of simplicity. We keep the asymmetry at the carbon level in reaction (3), as observed in Fig. 5b. Under this new scheme, the nodes of the network are now the carbon atoms of the metabolites. For example, A_1 and A_2 represent the first and second carbon atom of metabolite A. This also applies to metabolites B, C, D and B_{ext}. Arcs represent atom carbon fluxes inside a biochemical reaction. For example, in the first reaction, the first carbon of metabolite A (A_1) flows to the first carbon of metabolite B (B_1).

A new stoichiometric matrix for carbon fluxes can be defined and, therefore, the calculation of a new set of elementary solutions can be carried out. These elementary solutions, *i.e.* solutions with one degree of freedom, are here referred to as elementary carbon modes (ECMs). Fig. 5c shows the 5 ECMs emerging from Fig. 5b. ECMs can be linear (first, second, third and fourth ECMs) or cyclic (fifth). Note here that, as we have now unimolecular reactions, ECMs are simple paths or simple cycles, as defined in graph theory (Bondy et al., 1976). Though the number of ECMs is much higher than EFMs, the fact of being simple paths and cycles constitute an advantage from the computational point of view with respect to EFMs, whose complexity is beyond than graph theory (Acuña et al., 2009; Acuña et al., 2010).

In analogy with EFMs and metabolic networks, any steady-state carbon flux can be reconstructed as a linear combination of ECMs with non-negative coefficients. This is due to the fact that the steady-state carbon flux solution space, similarly to the steady-state reaction fluxes, forms a polyhedral cone (Wagner and Urbanczik, 2005). Any solution in a polyhedral cone can be reconstructed as a linear combination with non-negative coefficients of its underlying extreme rays (see Fig. 5d), which are precisely the elementary solutions, which in our case are ECMs.

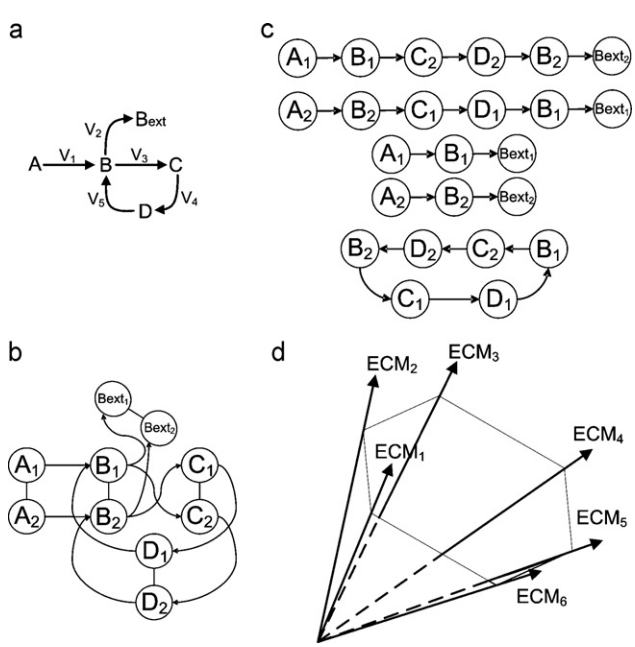

Fig. 5. (a) Example metabolic network; (b) example metabolic network at the carbon level; (c) resulting ECMs from (a); (d) carbon flux polyhedral cone and ECMs.

This property has two important consequences. Firstly, the set of ECMs capture any carbon (flux) movement in the metabolic network. For example, the first ECM defines the carbon flow from the first carbon of A to the second carbon of B_{ext}, which cannot be observed by using the EFMs approach proposed above. Secondly, ECMs combine linearly at the carbon level. This fact is interesting for the integration of ECMs and labeling data in a linear fashion.

In summary, the ECMs approach is a promising concept for the integration of tracer-based metabolomics data into metabolic networks. The next logical step is to relate ECMs with labeling data and reaction fluxes, giving so more practical application to ECMs. The relation of ECMs with reaction fluxes can be easily determined, just by considering that carbon fluxes belonging to one reaction flows at the same rate, namely the metabolic flux of such reaction. However, more research is needed to elucidate the relationship between tracer-based metabolomics data and ECMs.

4. Conclusions

In this article, we discussed whether EFMs and metabolomics data from ILEs can be integrated into a linear system of equations. Our discussion was based on the insight that the EFM isotopomer distribution is a structural property for given conditions. An extension of the model of Schmidt et al. (1997) was used to determine the EFM isotopomer distribution. A detailed analysis revealed that the linear combination of the isotopomer distribution of metabolites in EFMs is not sufficient to characterize carbon movements in ILEs. Indeed, the interaction of EFMs generates new carbon movements not considered by EFMs *per se*. Therefore, answering the question heading the article, EFMs do not combine linearly at the atomic level.

To be able to fully capture the carbon movements in a metabolic network using a linear model, the concept of EFM needs to be conveniently redefined. In this article, we introduced a novel definition of EFM at the carbon level, namely Elementary Carbon Modes (ECMs). We think that ECMs are a promising concept for the integration of tracer-based metabolomics data into metabolic networks in a linear fashion.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.biosystems.2011.04.005.

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